

A Dual-band 10-element Antenna with Good Gain and Efficiency for ISM and Industrial Automation Applications

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Abstract: This study presents a dual-band 10-element antenna with an inverted Π shaped defected ground structure (DGS) for applications in the Industrial, Scientific, and Medical (ISM) band and industrial automation. The proposed antenna consists of a 2×5 array structure, incorporating ten rectangular patch elements interconnected via sub-feeders. To achieve dual-band radiation and antenna miniaturization, a modified ground plane is implemented using an inverted Π shaped DGS. This design enhancement effectively optimizes the antenna's performance while making it well-suited for the targeted applications. The radiation patterns along with the reflection coefficients (S_{11}) are presented and analyzed in this study. The first resonance occurs at 2.59 GHz, with a corresponding return loss (S_{11}) of -34.95 dB, covering a frequency range of 2.19–2.96 GHz. The second resonance is observed at 9.20 GHz, with a return loss of -29.58 dB, spanning 8.84–9.50 GHz. These frequency bands are well-suited for applications in the ISM band and industrial automation, respectively. The proposed microstrip planar antenna array features a compact and structurally simple design, mounted on both sides of a Rogers RT 5880 substrate (dielectric constant, $\epsilon = 2.2$). The antenna has an overall dimension of $45 \times 35 \times 0.79$ mm³ and demonstrates a high maximum efficiency of 98.02%, along with favorable far-field characteristics. The design was validated by using both time domain and frequency domain analysis, confirming the antenna's suitability for the ISM and industrial automation applications.

Keywords: Antenna Array; Defected Ground Structure (DGS); ISM; Industrial Automation Applications

I. INTRODUCTION

In modern wireless communication systems, dual-band antennas are essential for ensuring efficient and reliable connectivity. By operating across two distinct frequency bands, they support diverse wireless standards, improve data transmission rates, and provide seamless, uninterrupted communication across various deployment scenarios [1]. In industrial automation, antennas enable wireless

communication for applications such as robotic control, machinery monitoring, and IoT integration, thereby enhancing operational efficiency and productivity [2]. The rise of autonomous vehicles has further increased demand for robust wireless technologies, as reliable data connectivity is critical for vehicle-to-vehicle and vehicle-to-infrastructure communication [3]. Antennas are also widely used across scientific and industrial domains, including remote sensing, electromagnetic heating, medical diagnostics, and automation. In the automotive sector, WiFi, Bluetooth, and X-band frequencies are commonly employed for connectivity and sensing applications [4]. Among the globally allocated ISM bands, the 2.4–2.5 GHz range is most widely used due to its availability and support for low-power technologies such as Bluetooth, WiFi, and ZigBee [5–6].

Recent advancements have spurred interest in miniature microstrip antennas due to their low-profile structure, cost-effectiveness, ease of fabrication, and integration with microwave circuits [7]. Despite these advantages, microstrip antennas typically suffer from limited efficiency and narrow bandwidth [8]. Defected Ground Structure (DGS) is a proven technique to overcome these limitations. By introducing intentional defects in the ground plane, DGS alters the inductance and capacitance of the transmission line, enabling multiband operation and improved bandwidth [9]. Additionally, DGS aids in antenna miniaturization and enhances radiation characteristics by controlling electromagnetic excitation and current distribution, making it a valuable design strategy in contemporary antenna engineering. A variety of antenna designs have been proposed for ISM band, automation and IoT-related applications, each optimized for specific use cases, performance metrics, and physical constraints [10].

In [7], a compact patch antenna tailored for IoT within the 2.45 GHz ISM band is presented. It occupies a footprint of $45 \times 50 \times 1.67$ mm³ and operates over the frequency spectrum of 2.42–2.47 GHz, achieving a peak gain of 5.8 dB. A dual-band antenna operating at center frequencies of 2.5

GHz and 6 GHz is introduced in [11]. It supports frequency ranges of approximately 2.4–3 GHz and 5.9–6.1 GHz, offering a gain of 3.8 dB and an efficiency of 90% within a size of $62 \times 62 \times 1.6 \text{ mm}^3$. In [12], a circular antenna based on characteristic mode analysis is proposed for ISM band applications. This antenna exhibits a -10 dB bandwidth from 2.38 to 2.49 GHz, a peak gain of 4.49 dB, and dimensions of $95 \times 75 \times 1.6 \text{ mm}^3$. The work in [13] presents a crescent-cut and extended circular patch antenna designed for the ISM band. With a footprint of $77 \times 77 \times 0.8 \text{ mm}^2$, it operates within the 2.35–2.5 GHz range and provides a peak gain below 6 dB. A flexible dual-band WiFi 6 antenna intended for wearable military applications is explored in [14]. It covers the 2.30–2.60 GHz and 5.00–5.30 GHz bands, with respective center frequencies at 2.45 GHz and 5.13 GHz. The antenna measures $50 \times 50 \times 0.99 \text{ mm}^3$ and delivers a peak gain of 6.384 dB with 73% efficiency. An enhanced rectenna for battery-free medical implants is detailed in [15], operating across 2.4–2.48 GHz and centered at 4.45 GHz. It offers a gain of 3.7 dB and an efficiency of 72%, within a size of $50 \times 50 \times 1.6 \text{ mm}^3$. In [16], a polarization-insensitive metamaterial absorber is proposed for ISM band use. It covers a wide frequency range from 1.94 to 2.98 GHz, demonstrates a maximum efficiency of 90%, and features a compact form factor of $55.7 \times 55.7 \times 1.6 \text{ mm}^3$. A passive beam-steering dielectric resonator antenna (DRA) for both static and dynamic subject applications is presented in [17]. It operates over 2.05–2.48 GHz, provides a peak gain of 5.5 dB, and achieves over 70% efficiency within dimensions of $75 \times 75 \times 1.6 \text{ mm}^3$.

An omnidirectional planar antenna in [18] employs parasitic directors and operates over 2.24–2.62 GHz, with a center frequency of 2.45 GHz. The design achieves a gain of 3.9 dB and an efficiency of 79.46%, with dimensions of $220.6 \times 220.6 \times 1.6 \text{ mm}^3$. A textile-based wearable antenna for ISM band applications is discussed in [19], operating from 2.4 to 2.6 GHz with a center frequency of 2.45 GHz. It exhibits a peak gain of 5.40 dB and occupies a size of $80 \times 84 \times 2 \text{ mm}^3$. In [20], a coplanar waveguide (CPW)-fed wearable antenna designed for Wireless Body Area Network (WBAN) is introduced. It operates in the 2.27–2.53 GHz range, centered at 2.39 GHz, and provides a gain between 2.07 and 3.04 dB with a size of $65 \times 65 \times 2.4 \text{ mm}^3$. A wearable meandered patch antenna developed for respiration monitoring is presented in [21], operating over 2.3–2.9 GHz and delivering a peak gain of 4.04 dB within a compact profile of $150 \times 53 \times 0.78 \text{ mm}^3$. Lastly, [22] describes a proximity-coupled wideband antenna optimized for RF energy harvesting. It functions within the 2.35–2.6 GHz range, offering a peak gain of 5.6 dB and an impressive efficiency of 92%, with dimensions of $60 \times 80 \times 2.4 \text{ mm}^3$. The reviewed works, while offering diverse designs for ISM and related bands, often involve trade-offs between size, gain, bandwidth, and efficiency, with many antennas lacking optimal compactness or wideband performance for wearable and implantable applications. Additionally, several designs show limited integration flexibility for conformal or multi-functional use, which restricts their practical deployment in dynamic IoT and biomedical environments.

In this work, a dual-band 10-element antenna with operating frequencies of 2.19–2.96 and 8.84–9.50 GHz has been presented. A 2×5 array structure has been developed in this proposed antenna for ISM and industrial automation applications utilizing an inverted Π -shaped slot on the back

part of the substrate, and this antenna operates at 2.59 GHz and 9.2 GHz. The antenna's peak gain is 6.69 dB, with VSWRs of 1.03 and 1.06, and its highest radiation efficiency is 98.02%. The organization of this paper is as follows: Section II details the design, configuration and evolution of the antenna. Section III presents the results and analysis obtained from simulations. Finally, Section IV provides the conclusion.

II. STRUCTURE AND ANTENNA EVOLUTION

The design and simulation of the proposed dual-band 10-element antenna with an inverted Π shaped DGS for ISM and industrial automation applications were conducted using CST Microwave Studio (CST-MWS) to ensure precise optimization. This study presents a detailed examination of the antenna's design methodology, structural configuration, and geometric parameters. Figs 1(a) and (b) illustrate the top view, back view of the antenna, highlighting key components. The initial dimensions were established theoretically and subsequently optimized via parametric adjustments to enhance performance metrics, including impedance matching, gain, and bandwidth. The antenna is constructed on a Rogers RT5880 (lossy) substrate, characterized by a dielectric constant of 2.2 and a loss tangent of 0.0009. The substrate has a thickness h of 0.79 mm and an overall size ($L \times W$) of $45 \times 35 \text{ mm}^2$. The radiating elements and feeding structures utilize annealed copper with a thickness of 0.035 mm, ensuring high conductivity and minimal losses. The feeding mechanism integrates a rectangular feed line, connecting strips, sub-feeders, and a 2×5 array structure, all contributing to optimized impedance matching, current flow, and radiation efficiency. The stepwise design process is as follows:

1. A Rogers RT5880 substrate with dimensions of $45 \times 35 \times 0.79 \text{ mm}^3$ is selected as the substrate material. A rectangular feed line ($a \times b$), measuring $10 \times 2.4 \text{ mm}^2$, is positioned on the front surface, as shown in Fig. 1(a).
2. Two connecting strips ($c \times d$), each measuring $10.8 \times 2 \text{ mm}^2$, are symmetrically attached to the upper left and right sections of the feed line to facilitate efficient signal propagation.
3. A 2×5 array structure is constructed by arranging two parallel branches that are connected with main feeder ($a \times b$) through the connecting feeders ($c \times d$). Both left and right each of the parallel branches contains five rectangular patch elements ($e \times f$) with dimensions of $4 \times 5 \text{ mm}^2$, situated in the middle section on the substrate. These patches are interconnected via square-shaped sub-feeders, each with a length of 2 mm, promoting uniform current distribution and enhancing radiation properties.
4. To improve impedance matching and bandwidth, an inverted Π shaped DGS is introduced on the back side of the antenna. A ground plane ($j \times k$) of dimensions $45 \times 35 \text{ mm}^2$ is strategically placed at the back section of the substrate, as depicted in Fig. 1(b). Then one inverted Π shaped slot has been created in the middle section of the ground plane. To create this structure, first a horizontal top bar ($n \times m$) with dimensions of $35 \times 4.5 \text{ mm}^2$ has been created on the lower portion of the ground plane. Next, two vertical legs ($p \times q$) of dimensions $30 \times 6 \text{ mm}^2$ have been constructed on the left and right corners of the ground plane. The distance (r) between two vertical legs is 16 mm. These two vertical legs are connected with the

horizontal top bar. In this way, an inverted Π shaped DGS has been created on the ground plane. This modification alters surface current flow, contributing to the effective formation of the DGS. The entire design evolution process has been summarized in Fig. 2.

Following extensive simulation and iterative refinement, the final antenna design achieves excellent performance in terms of gain, impedance matching, and bandwidth, making it a strong candidate for ISM and industrial automation applications. A comprehensive list of structural parameters for the proposed antenna is provided in Table I.

TABLE I LIST OF PARAMETERS

Description	Symbol	Value (mm)
Length of feeder	a	10
Width of feeder	b	2.4
Length of the connecting strip	c	10.8
Width of the connecting strip	d	2
Length of the rectangular patch element	e	4
Width of the rectangular patch element	f	5
Length of the sub-feeder	g	2
Length of the ground plane	j	45
Width of the ground plane	k	35
Length of the horizontal top bar	n	35
Width of the horizontal top Bar	m	4.5
Length of the vertical legs	p	30
Width of the vertical legs	q	6
Distance between two vertical legs	r	16

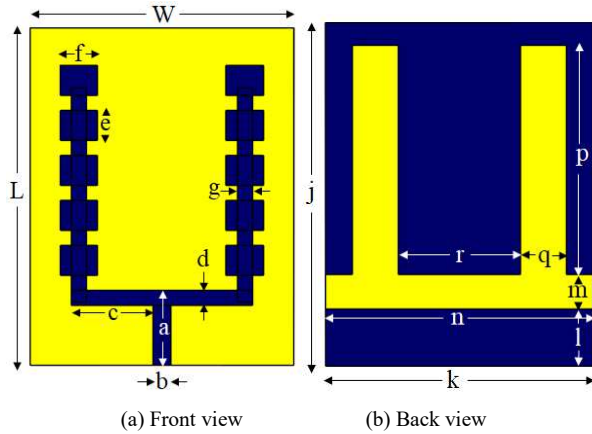


Fig.1. Dual-band 10-element antenna with an inverted Π shaped DGS

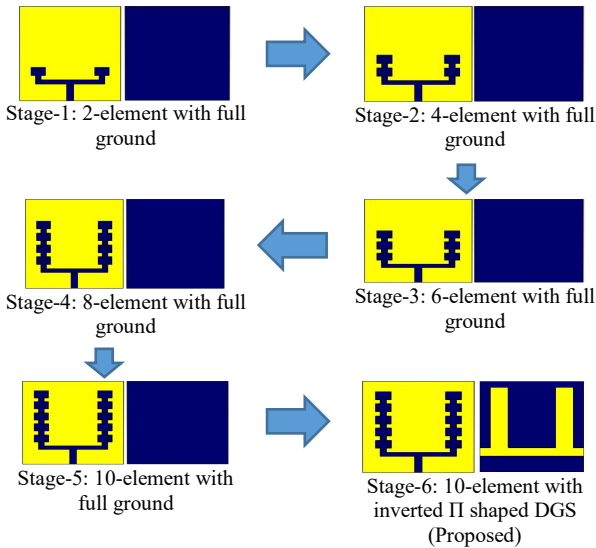


Fig.2. Design evolution steps of dual-band 10-element antenna with an inverted Π shaped DGS

III. RESULTS AND ANALYSIS

All the estimated performance of the designed dual-band 10-element antenna with an inverted Π shaped DGS for ISM and industrial automation applications is presented in this section. The antenna's characteristics are analyzed through both time domain and frequency domain analysis, with its radiation properties evaluated based on the configurations outlined in the previous section. The CST-MWS suite, high-performance 3D electromagnetic (EM) simulation software, is utilized for the design, analysis, and optimization of the proposed antenna. In patch antennas, the reflection coefficient is utilized to assess the impedance mismatch among the antenna and the feeding network and is closely related to return loss. The antenna's simulated reflection coefficients for all the six stages are shown in Fig. 3. The antenna has resonances at 2.59 GHz and 9.2 GHz, according to the data from the CST software. The corresponding return loss is -34.95 and -29.58 dB, respectively. This antenna operates from 2.19 to 2.96 GHz and from 8.84 to 9.50 GHz. Performance is facilitated and signal power is increased by the 10-element array design. It covers ISM and industrial automation band applications.

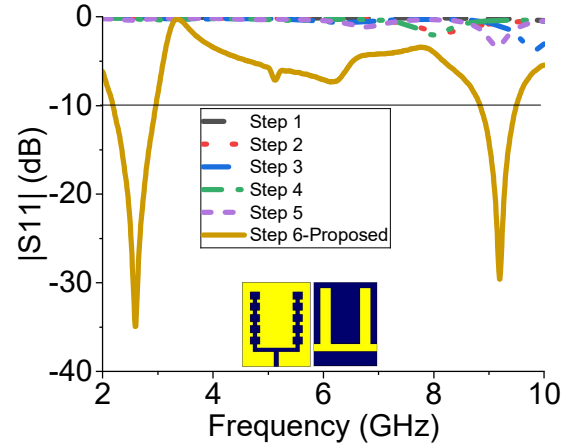


Fig. 3. Reflection coefficient variation of all the evolution steps

Gain is a crucial parameter in patch antennas, reflecting their ability to focus and direct emitted energy in a specific direction. The gain variation of all the evolution steps is presented in Fig. 4 which indicates that the proposed one shows best gain profile in the lower operating frequency band as well as suitable gain profile in the higher operating band. The operating frequency ranges are represented by the highlighted areas in Fig. 5, which illustrate the linear gains and directivities. For an operating frequency range of 2.19-2.96 GHz, the gain ranges from 2.33 dB to 3.77 dB, while directivity varies between 2.92 dBi and 3.86 dBi. In the second resonant frequency range of 8.84-9.50 GHz, the gain ranges from 5.69 dB to 6.69 dB, whereas the directivity varies between 6.19 dBi and 7.35 dBi. At the center frequency of 2.59 GHz, the antenna demonstrates a gain of 2.23 dB and a directivity of 3.14 dBi as depicted in Fig. 6, making it an excellent choice for ISM band applications systems. At the second center frequency of 9.20 GHz, the gain and directivity are 6.44 dB and 7.30 dBi, respectively as shown in Fig. 7. It achieves a maximum directivity of 7.35 dBi and a maximum gain of 6.69 dB at 8.84 GHz.

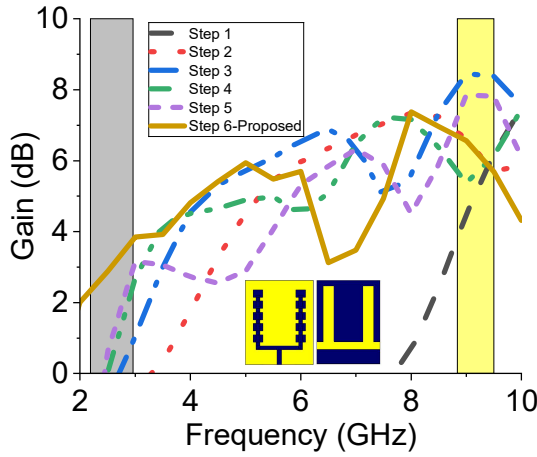


Fig. 4. Gain variation of all the evolution steps

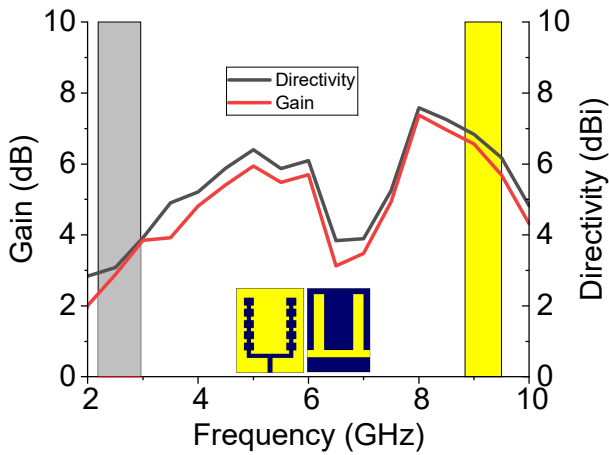


Fig. 5. Linear gain and directivity of the dual-band 10-element antenna with an inverted II shaped DGS

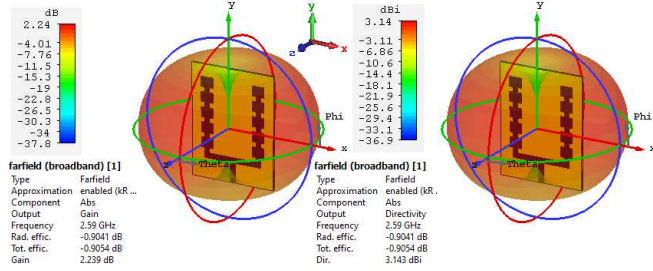


Fig. 6. Gain and Directivity of the dual-band 10-element antenna with an inverted II shaped DGS at 2.59 GHz

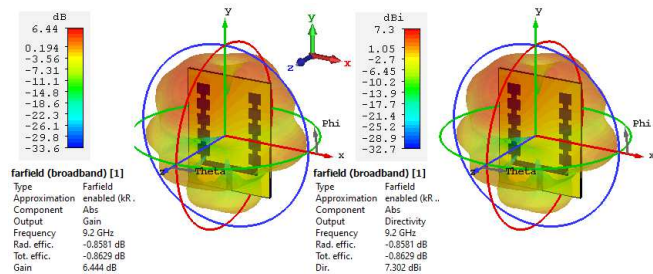


Fig. 7. Gain and Directivity of the dual-band 10-element antenna with an inverted II shaped DGS at 9.20 GHz

A real part that is close to fifty and an imaginary part that is close to zero are required for effective antennas. Fig. 8 displays the Z parameter for the suggested antenna. The Z

parameter for this antenna has real parts of 51.09 and 51.98 at the center frequencies of 2.59 GHz and 9.20 GHz, respectively. This antenna's imaginary parts are extremely near to zero at both operational frequencies.

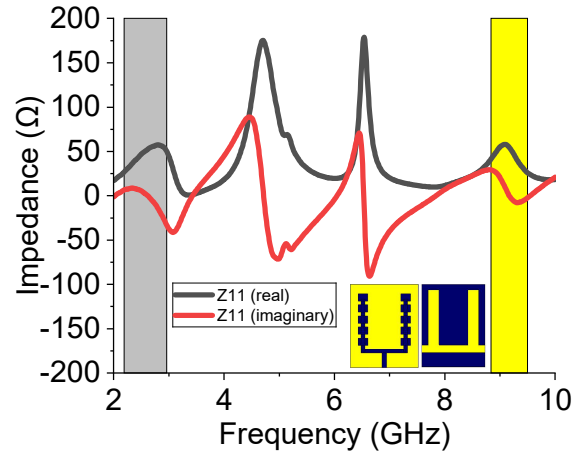


Fig. 8. Z-parameters of the dual-band 10-element antenna with an inverted II shaped DGS

The efficiency of patch antennas refers to their ability to effectively convert input power into radiated power. Fig. 9 illustrates the radiation efficiency and VSWR of a dual-band 10-element antenna with a unique defected ground structure. The efficiency at the center frequency of 2.59 GHz is 95.99%, which is considered a good result. In the lower frequency range of 2.19-2.96 GHz, the efficiency varies between 87.50% and 98.02%. In the upper frequency range of 8.84-9.50 GHz, the efficiency ranges from 89.20% (at upper cutoff) to 93.71% (at lower cutoff). At the second resonance point of 9.20 GHz, it possesses efficiency of 91.95%. An accurate indicator of how well an antenna is impedance-matched to its feedline is VSWR. A VSWR of 1 represents a higher effective power throughput and minimal reflection. The proposed 10-element antenna achieves an excellent VSWR of 1.03 at its center frequency of 2.59 GHz. This value meets the primary design requirement of maintaining a VSWR below 2. At the lower cutoff frequency of 2.19 GHz, the VSWR is 1.91, while at the upper cutoff frequency of 2.96 GHz, it is 1.90. This indicates that the antenna effectively operates within the 2.19 GHz to 2.96 GHz range, maintaining a VSWR below 2. Similarly, at its second center frequency of 9.20 GHz, the proposed antenna exhibits a favorable VSWR of 1.06, which remains below 2. At the lower cutoff frequency of 8.84 GHz, the VSWR is 1.92, while at the upper cutoff frequency of 9.50 GHz, it is 1.93. This confirms that the antenna covers the frequency range of 8.84 GHz to 9.50 GHz while maintaining a VSWR below 2. The surface current distribution scenario has been depicted in Fig. 10 at both center frequencies. For 2.59 GHz, maximum current is 67.46 A/m whereas for 9.20 GHz, it is 66.50 A/m. Electric and magnetic fields in polar form has been depicted in Fig. 11 for $\phi = 0$ and 90 degree which indicates that the designed dual band antenna is suitable for ISM (first operating band) and industrial automation (2nd operating band) applications. An extensive comparison table II vividly presents the strength of the proposed work compared to some recently published relevant works.

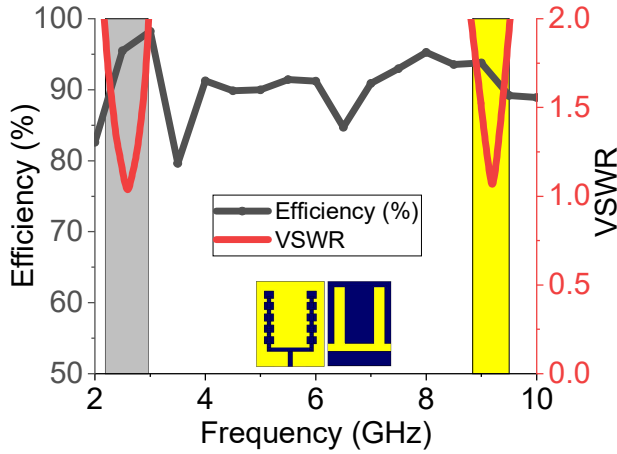


Fig. 9. Efficiency and VSWR of the dual-band 10-element antenna with an inverted II shaped DGS

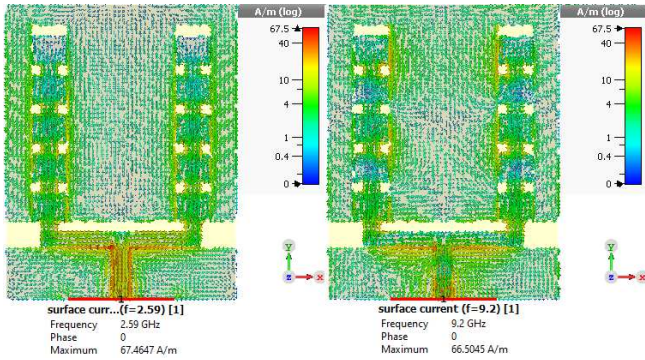


Fig. 10. Surface current of the dual-band 10-element antenna with an inverted II shaped DGS

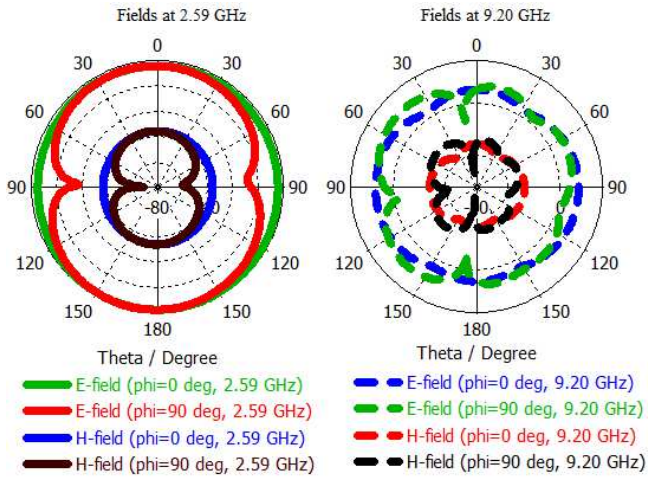


Fig. 11. Fields of the dual-band 10-element antenna with an inverted II shaped DGS

IV. CONCLUSION

In this study, a compact dual-band 10-element ($45 \times 35 \times 0.79$ mm³) microstrip array antenna has been designed and validated for applications in ISM band and industrial automation. The proposed antenna, featuring a 2×5 array structure with interconnected rectangular patch elements and an inverted II shaped DGS. The proposed antenna exhibits a gain range of 2.33 to 6.69 dB. The antenna demonstrating resonance at 2.59 GHz and 9.20 GHz with wide impedance bandwidths, a high maximum efficiency of 98.02%, and favorable far-field characteristics. The antenna's compact

size, structural simplicity, and high-performance metrics make it a strong candidate for modern industrial and medical wireless communication systems. Future work may explore further miniaturization, bandwidth enhancement, and integration with advanced communication technologies to expand its applicability.

REFERENCES

- [1] A. S. Burra and B. Roy, "Design and Evaluation of a Compact Unified Hexagonal Dual-Band Wearable Patch Antenna," *Heliyon*, vol. 11, e42868, pp. 1-16, 2025.
- [2] K. A. Manjunatha and V. Agarwal, "Multi-band heterogeneous wireless network architecture for industrial automation: A techno-economic analysis," *Wireless Pers. Commun.*, vol. 123, no. 4, pp. 3555-3573, 2022.
- [3] T.L. Chiu, O. Pajona, J. Kyllonen, L. Desclos, L. Huitema, and T. Monediere, "Active steering dielectric resonator antenna for automotive," in *Proc. Int. Workshop Antenna Technol. (iWAT)*, Athens, Greece, 2017, pp. 343-346, doi: 10.1109/iWAT.2017.7915396.
- [4] A.P. Buta, A.-M. Silaghi, A. De Sabata, and L. Matekovits, "Multiple-Notch Frequency Selective Surface for Automotive Applications," in *Proc. Int. Conf. Commun. (COMM)*, Bucharest, Romania, 2020, pp. 439-442, doi: 10.1109/COMM48946.2020.9142001.
- [5] A. El Ansari *et al.*, "Conception and Realization of a Wideband Directional Dual-Beam Phased MIMO Array Antenna with Hybrid Coupler for ISM Band Utilizations," *Progress In Electromagnetics Research M*, vol. 132, pp. 39-48, 2025.
- [6] S. B. Naik *et al.*, "A 1 V Double-Balanced Mixer for 2.4-2.5 GHz ISM Band Applications," in *Proc. Int. Conf. VLSI Design and Embedded Syst. (VLSID)*, Guwahati, India, 2021, pp. 252-257, doi: 10.1109/VLSID51830.2021.00048.
- [7] M. Harrou *et al.*, "Design of Patch antenna with Various shapes and various feeding techniques for IoT in the 2.45 GHz ISM Band," in *ITM Web Conf.*, vol. 69, p. 04015, 2024.
- [8] S. Angadi *et al.*, "Meta-atom loaded circularly polarized triple band patch antenna for Wi-Fi, ISM and X-band communications," *Heliyon*, vol. 10, no. 7, 2024.
- [9] N. Akshitha *et al.*, "Gain and bandwidth improvement of microstrip patch antenna using defected ground structure (DGS) and copper reflector," in *Hybrid and Advanced Technologies*, CRC Press, pp. 192-197, 2025.
- [10] A. A. Megahed, A. H. Hussein, A. J. A. Al-Gburi, and R. H. Elabd, "Compact Wideband Antenna Array with DGS-Based Metamaterial for Efficient Smartphone Communication and SAR Reduction," *Prog. Electromagn. Res. B*, vol. 110, pp. 5-25, 2025.
- [11] E.D.K. Ruby *et al.*, "Compact circularly polarized dual band antenna with modified patch for ISM and V2X communication systems," *Sci. Rep.*, vol. 15, no. 1, p. 5885, 2025.
- [12] Y. Sabila *et al.*, "Analysis of Radiation Structure of Circular Microstrip Antenna using Characteristic Mode Analysis for ISM Band," *Elkomika*, vol. 11, no. 1, pp. 100, 2023.
- [13] M. M. B. Mojumdar, R. I. Rabbi, D. Das, and M. A. Hossain, "Crescent-cut and Extended Circular Patch Circularly Polarized Microstrip Antenna for ISM Band Wireless Applications," in *Proc. Int. Conf. Adv. Electr. Electron. Eng.*, Gazipur, Bangladesh, 2024, pp. 1-6, doi: 10.1109/ICAEEE62219.2024.10561647.
- [14] P. Harshini, V. Rathinasamy, and D. Thangarasu, "A Flexible Wi-Fi 6 Antenna with low SAR for Wearable Military Applications," in *Proc. Int. Conf. Signal Process., Comput., Electron., Power Telecommun.*, Karaikal, India, 2024, pp. 1-5, doi: 10.1109/ICoNSCEPT61884.2024.10627824.
- [15] Y. Bouchoucha, D. Omri, and T. Aguil, "Efficient ISM Band Rectenna for RF Energy Harvesting in IoT Healthcare Devices," in *Proc. Int. Conf. Commun., Comput., Cybersecurity, Informatics*, Chongqing, China, 2023, pp. 1-8, doi: 10.1109/CCCIS58712.2023.10290775.
- [16] A. A. G. Amer *et al.*, "A Wide-Angle, Polarization-Insensitive, Wideband Metamaterial Absorber With Lumped Resistor Loading for

- ISM Band Applications," *IEEE Access*, vol. 12, pp. 42629–42641, 2024, doi: 10.1109/ACCESS.2023.3345659.
- [17] V. R. Gudivada, Y. Huang, and E. L. Bennett, "A passive beamforming hemispherical liquid antenna using a simple metal ball," *IET Microw. Antennas Propag.*, vol. 18, no. 9, pp. 681–690, 2024.
- [18] P. Kumar *et al.*, "High - gain horizontally polarised omnidirectional planar antenna using parasitic directors for wireless sensor networks," *Int. J. Commun. Syst.*, vol. 37, no. 6, e5697, 2024.
- [19] A. Marimganti *et al.*, "Wearable Textile Antenna for On-Body Communication," in *Proc. Int. Conf. Comput. Intell. Inf., Security Commun. Appl.*, Bengaluru, India, 2023, pp. 1–6, doi: 10.1109/CIISCA59740.2023.00011.
- [20] T. Sheikh *et al.*, "Wearable CPW Antenna for WBAN Applications," in *Proc. Int. Conf. Intell. Technol.*, Hubli, India, 2023, pp. 1–5, doi: 10.1109/CONIT59222.2023.10205758.
- [21] M. H. Fathurridha *et al.*, "Wearable Antenna with DGS Technique for Enhanced Gain in Respiration Detection Applications," in *Proc. IEEE Asia-Pacific Microw. Conf.*, Bali, Indonesia, 2024, pp. 1108–1110, doi: 10.1109/APMC60911.2024.10867634.
- [22] B. R. Behera, N. K. Sahu, and S. K. Mishra, "Proximity Coupled Wideband Rectangular Microstrip Antenna for RF Energy Harvesting," in *Proc. Int. Conf. Circuits, Power Intell. Syst.*, Bhubaneswar, India, 2023, pp. 1–4, doi: 10.1109/CCPIS59145.2023.10292012.

TABLE II COMPARISON TABLE

Index	Reference No.													This work
	7	11	12	13	14	15	16	17	18	19	20	21	22	
Year	2024	2025	2023	2024	2024	2023	2023	2024	2024	2023	2023	2024	2023	2025
Size (L×W×h) mm ³	45×50×1.67	62×62×1.6	95×75×1.6	77×77×0.8	50×50×0.99	50×50×1.6	55.7×55.7×1.6	75×75×1.6	220.6×220.6×1.6	80×84×2	65×65×2.4	150×53×0.78	60×80×2.4	45×35×0.79
Volume	3757.5	6150.4	11400	4743.2	2475	4000	4963.98	9000	77862.98	13440	10140	6201	11520	1244.25
Substrate material	Rogers RT/Duroid 5880	FR4	FR4	Teflon	Polyester	Rogers RT 5880	FR4	Teflon (PTFE)	FR4	Denim	Denim	Textile	Rogers RT 5880	Rogers RT 5880
Operating Frequency Range (GHz)	2.42 - 2.47	~ 2.4-3 and ~ 5.9-6.1	2.38-2.49	2.35-2.5	~ 2.30 - 2.60 and ~ 5-5.3	2.4-2.48	1.94 to 2.98	2.05-2.48	2.24 – 2.62	~ 2.4-2.6	2.27-2.53	~ 2.3-2.9	~ 2.35-2.6	2.19-2.96, 8.84-9.50
Centre frequency (GHz)	2.45	2.5 and 6	2.43	2.45	2.45 and 5.13	2.45	2.4	2.4	2.45	2.45	2.39	2.376	2.4	2.59, 9.20
Return loss (dB) at centre frequency	-37.53	-30 and -16	-23.66	-31.47	~ -12 and ~ -14	-40	~ -45	~ -29	~ -33	-25.36	-34	-32.36	~ -18	-34.95, -29.58
Gain (dB)	5.8	3.8	4.49	< 6	6.384	3.7		5.5	3.9	5.40	2.07-3.04	4.04	5.6	2.33-6.69
-10 dB BW (GHz)	0.05	0.6, 0.2	0.11	0.15	0.3, 0.3	0.08	1.04	0.43	0.38	0.2	0.26	0.6	0.25	0.77, 0.66
Max. radiation efficiency (%)	-	90	-	-	73	72	90	>70	79.46	-	-	-	92	98.02
Application	ISM	Next-gen technology	ISM	ISM	Military	RF Energy Harvesting	ISM	ISM	Wireless sensor networks	ISM	WBAN	Respiration Detection	RF Energy Harvesting	ISM and industrial automation